

# Graph-Enhanced Causal Reinforcement Learning for Proactive Customer Retention

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A thesis submitted to the Department of Computer Science and Engineering  
in partial fulfillment of the requirements for the degree of  
B.Sc. in Computer Science and Engineering

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# Declaration

It is hereby declared that

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2. The thesis does not contain material previously published or written by a third party, except where this is appropriately cited through full and accurate referencing.
3. The thesis does not contain material which has been accepted, or submitted, for any other degree or diploma at a university or other institution.
4. We have acknowledged all main sources of help.

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# Abstract

Customer retention is a critical business challenge, as acquiring new customers is significantly more expensive than retaining existing ones. Conventional machine learning approaches to churn management focus on predicting attrition probability but fail to identify which customers will actually respond to retention interventions—a disconnect known as retention futility. This literature review synthesises research across customer retention analytics, causal inference and uplift modelling, offline reinforcement learning, graph neural network architectures, graph-based causal inference, and temporal graph methods to establish the theoretical and methodological foundations for prescriptive retention. The review reveals that causal uplift modelling enables actionable segmentation of customers into Persuadables, Sure Things, Lost Causes, and Sleeping Dogs, while graph neural networks capture relational structure and network confounding that tabular approaches miss. Offline reinforcement learning provides frameworks for learning intervention policies from historical data without the risks of live experimentation. However, critical gaps persist: existing methods operate in methodological silos, cold-start scenarios remain inadequately addressed, validation relies on limited benchmarks, and the Sleeping Dog phenomenon lacks explicit algorithmic safeguards. These findings motivate the development of a Graph-Enhanced Causal Reinforcement Learning framework that leverages graph embeddings as deconfounded state representations, estimates heterogeneous treatment effects through uplift modelling, and optimises a conservative offline policy via Batch-Constrained Q-Learning. The proposed framework bridges the predictive–prescriptive gap in retention analytics by grounding intervention decisions in causal mechanisms rather than correlational patterns.

**Keywords:** Customer Churn Prediction, Graph Neural Networks, Causal Inference, Heterogeneous Treatment Effects, Uplift Modeling, Off-Policy Evaluation, Offline Reinforcement Learning, LightGCN, Batch-Constrained Q-Learning, Network Confounding, Dynamic Graph Neural Networks, Open Bandit Dataset.

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# Chapter 1

## Introduction

### 1.1 Background

Predictive analytics is transforming Customer Relationship Management (CRM) from reactive churn identification towards prescriptive frameworks that not only predict potential churners but also establish concrete and effective retention measures. Conventional churn forecasting models, including classical machine learning methods such as Random Forest and XGBoost, tend to confuse correlation with causation and are therefore limited in their capacity to guide actionable interventions. Research has consistently demonstrated that a modest increase in customer retention rates can lead to substantial profit improvements [1], underscoring the enormous economic value of effective retention strategies.

To counter these limitations, the proposed study adopts Uplift Modelling to approximate the Individual Treatment Effect (ITE) of each customer and classify them into actionable segments: Persuadables (who respond positively to intervention), Sure Things (who stay regardless), Lost Causes (who churn despite intervention), and Sleeping Dogs (who churn because of intervention). The problem is formulated as a Batch-Constrained Contextual Bandit framework, applied to safely optimise retention policies using Batch-Constrained Q-Learning (BCQ) from historical data. To overcome data sparsity and improve causal state representations, LightGCN, a simplified Graph Neural Network architecture, is employed to model user-item relationships and capture latent preferences while alleviating the cold-start problem. Finally, Inverse Propensity Weighting (IPW) and Doubly Robust Off-Policy Evaluation (OPE) estimators are applied to validate model performance on historical data, ensuring sound, data-driven customer retention decision-making without the risks associated with live experimentation.

### 1.2 Rationale of the Study or Motivation

This study is motivated by the imperative to advance customer retention management from reactive churn prediction to proactive, causally grounded intervention design. Several intersecting challenges in current retention practice and methodology drive this research.

First, classic churn prediction models achieve high classification accuracy but provide no guidance on which interventions will be effective for which customer segments. Empirical evidence from randomised field experiments has demonstrated



that customers with the highest predicted churn probability often exhibit the lowest response to retention campaigns—a phenomenon termed “retention futility.” This finding challenges the fundamental assumption underlying most CRM systems and motivates the adoption of uplift modelling, which approximates the Conditional Average Treatment Effect (CATE) to identify Persuadable customers who genuinely benefit from targeted action.

Second, real customer retention data suffers from confounding, data sparsity, and selection bias. Offline Reinforcement Learning offers the promise of learning optimal policies from historical interaction logs without costly online experimentation, but standard offline RL algorithms suffer from distributional shift and extrapolation error when evaluating actions that were rarely or never taken under the behaviour policy. Methods such as Batch-Constrained Q-Learning and Conservative Q-Learning address these challenges through policy constraints and conservative value estimation, respectively.

Third, user–item relationships contain rich collaborative signals that tabular features alone cannot capture. Graph Neural Networks, and LightGCN in particular, learn structural representations from interaction graphs that can serve as deconfounded state representations for the reinforcement learning agent. This graph-based approach also mitigates the cold-start problem by propagating information from well-observed nodes to newly arriving customers.

Finally, validating learned policies offline requires principled methods that account for the mismatch between the behaviour policy and the target policy. Off-Policy Evaluation techniques such as the Doubly Robust estimator and the Open Bandit Pipeline provide the evaluation infrastructure needed to benchmark prescriptive policies safely and reproducibly on real-world logged data.

## 1.3 Problem Statement

Customer retention is considerably less expensive than customer acquisition, yet most CRM systems continue to prioritise churn prediction—identification rather than prevention—creating a fundamental disconnect between analytical insight and practical action. Conventional predictive models estimate churn probability through correlational patterns, leading to ineffective targeting and often counterproductive marketing campaigns. Although Reinforcement Learning has the potential to derive dynamic retention strategies, offline learning environments present confounding, extrapolation errors, and poor generalisation. Current models also fail to capture relational patterns between users and products, and the absence of reliable Off-Policy Evaluation precludes safe policy validation. This research addresses four specific sub-problems:

1. **Predictive–Prescriptive Gap in Retention Strategy.** Traditional churn prediction models focus on forecasting customer attrition but do not identify the causal effect of interventions on behaviour. This limitation leads to mis-targeting and methodological debt, where marketing efforts unintentionally decrease retention effectiveness.
2. **Confounding and Bias in Offline Reinforcement Learning.** Reinforcement Learning for retention optimisation is hampered by hidden biases in

historical data. Embedding causal inference into RL is necessary to correct spurious correlations and enable reliable counterfactual reasoning.

3. **Representation Limitations and Data Sparsity.** Current retention systems operate with impoverished state representations that ignore user–item relational structure. Graph Neural Networks model the collaborative interaction graph to enrich causal state representations and address cold-start and sparsity problems.
4. **Validation and Evaluation Challenges in Policy Assessment.** Prescriptive policies are difficult to evaluate owing to unobservable counterfactual outcomes and the instability of conventional Off-Policy Evaluation techniques such as importance sampling. Stronger estimators are required to provide safe and reliable validation on real-world datasets.

This study proposes a Graph-Enhanced Causal Reinforcement Learning model for prescriptive customer retention. The model uses causal uplift estimation to identify customers most responsive to interventions, graph-based learning to overcome data sparsity and cold-start problems, and causal inference integrated into offline RL to address confounding. Sound Off-Policy Evaluation tools ensure policy safety and efficacy, creating a cohesive, data-driven system that transforms CRM from prediction-based to causally informed decision-making.

## 1.4 Objective

The overarching objective of this study is to design, develop, and validate a Graph-Enhanced Causal Reinforcement Learning (CRL) framework that integrates causal inference and graph-based representation learning to enable proactive customer retention. The framework aims to optimise retention interventions that maximise Customer Lifetime Value (CLV) while minimising risks from hidden confounding and unintended churn effects. This objective is decomposed into four targeted research objectives:

1. **Bridge the Predictive–Prescriptive Gap.** Transition the retention paradigm from churn probability estimation to uplift modelling, enabling prediction of causal effects of individual interventions and identification of Persuadable customers—those who respond favourably to targeted action—while avoiding harmful contact with Sleeping Dogs.
2. **Overcome Structural Blindness and Data Sparsity.** Introduce Graph Neural Networks, specifically LightGCN, to learn user–item interaction structure as graph embeddings, enabling the reinforcement learning agent to leverage collaborative signals and contextual patterns while addressing cold-start and sparsity challenges.
3. **Optimise Decisions Under Confounding.** Apply Offline Reinforcement Learning integrated with causal inference to eliminate biases arising from hidden confounding factors in historical data, ensuring that policy decisions are grounded in true causal relationships rather than spurious correlations.

4. **Validate Through Robust Off-Policy Evaluation.** Develop and benchmark Off-Policy Evaluation techniques, including Inverse Propensity Weighting and Doubly Robust estimation, to ensure the safety, stability, and real-world reliability of the proposed framework without live experimentation.

## 1.5 Methodology in Brief

The research methodology follows a multi-stage design that integrates graph-based representation learning, causal inference, and offline reinforcement learning into a unified prescriptive customer retention framework. The methodology proceeds through five interconnected stages.

### Stage 1: Literature and Problem Analysis

A comprehensive review of existing research across customer retention and churn prediction, causal inference and uplift modelling, offline reinforcement learning and contextual bandits, core graph neural network architectures, graph-based causal inference, and dynamic temporal graph models identifies the weaknesses of current approaches. This analysis demonstrates that purely correlational predictive systems have critical flaws, that offline data contain biases from hidden confounding, and that adequate evaluation procedures for safe policy deployment are lacking. The findings of this stage inform the conceptual foundation and technical direction of the proposed framework.

### Stage 2: Framework Design

The research develops a unified Graph-Enhanced Causal Reinforcement Learning (CRL) model combining three components: (a) LightGCN for learning graph-based state representations from user-item interaction data, (b) causal uplift estimation for identifying heterogeneous treatment effects across customer segments, and (c) Batch-Constrained Q-Learning (BCQ) for learning safe retention policies from logged historical data. The framework is designed to be causally interpretable, computationally efficient, and deployment-ready.

### Stage 3: Data, Training, and Backtesting

The Open Bandit Dataset (OBD), provided by ZOZO, Inc., serves as the primary data source. It contains large-scale logged interaction data with user contexts, actions (interventions), outcomes (click indicators), and propensity scores from both random and Thompson Sampling behaviour policies. Implementation proceeds through: (i) constructing user-item bipartite graphs from interaction logs, (ii) training LightGCN to produce dense graph embeddings as contextual state representations, (iii) estimating causal treatment effects using uplift modelling techniques, and (iv) training the BCQ bandit agent using graph embeddings as states, intervention IDs as actions, and click indicators as rewards. Backtesting on historical data enables iterative policy optimisation without the cost or risk of online experimentation.

## Stage 4: Evaluation

The evaluation stage combines causal and operational metrics to assess model validity and practical impact. Key metrics include the Conditional Average Treatment Effect (CATE) for causal uplift measurement, expected policy value via Off-Policy Evaluation, and retention uplift percentage for practical improvement over baseline policies. Off-Policy Evaluation uses the Doubly Robust estimator and Inverse Propensity Weighting, implemented through the Open Bandit Pipeline library. The learned policy is compared against the Random baseline and the Thompson Sampling baseline from the OBD, along with ablation variants (BCQ-only without GNN, LightGCN-only without causal bandits).

## Stage 5: Robustness, Scalability, and Generalisation

Built-in regularisation mechanisms and policy constraints prevent overfitting and bias, while graph-based modelling supports efficient learning on large, sparse datasets. The modular system design allows adaptation to different data domains and customer segments. Compliance and deployment readiness are verified through comprehensive Off-Policy Evaluation on logged data, ensuring that the final system conforms to ethical and operational standards while delivering accurate, fair, and transparent decision support.

## 1.6 Scopes and Challenges

### Scopes

- **Domain and Setting:** The study focuses on customer churn and retention decision-making in subscription and relationship-based digital businesses, rather than addressing all potential business analytics problems.
- **Data and Population:** The research operates on historical interaction records, transactional and behavioural logs, and graphical customer–item relationships drawn from the Open Bandit Dataset. It targets clientele in a specific segment of e-commerce markets, not the entirety of potential demographics or sectors.
- **Methods and Modelling:** The methodology encompasses graph representation learning (LightGCN), causal inference (uplift modelling), and offline reinforcement learning (BCQ). The study does not attempt an exhaustive comparison with all machine learning or deep learning architectures but builds and evaluates one unified framework within this triad.
- **Activities and Goals:** The primary aim is prescriptive retention policy learning—identifying which customers to target, when to act, and how to personalise retention activities. The project does not intend to create a general-purpose customer analytics system or an operational real-time production system.
- **Evaluation Environment:** All evaluation is conducted in an offline setting using Off-Policy Evaluation on logged historical data. No live A/B tests or on-

line deployments are performed. Empirical results report offline policy quality and safety, but not direct measurable impact on production business outcomes.

## Challenges

- **Data Issues:** Limited access to large, diverse, and representative data may introduce sampling bias. Interaction logs and graph structures are inherently noisy, may contain missing edges or incomplete records, and may not fully capture all factors influencing churn and treatment response.
- **Limitations of Causal Inference:** Unobserved confounders and latent variables may affect treatment assignment and outcomes, violating core assumptions of causal identification. The model relies on assumptions such as ignorability and overlap, whose validity in real business data is inherently approximate, potentially weakening causal conclusions.
- **Limitations of Offline RL:** Learned policies are constrained by the recorded behaviour policy and historical action distributions. Distributional drift between past and target policies, together with extrapolation to rarely observed actions, means that strong offline performance does not necessarily translate directly to online effectiveness.
- **Computational and Resource Constraints:** Hardware and time constraints limit the scale of graphs, the size of the action space, and the breadth of hyperparameter tuning and baseline comparisons. These restrictions may preclude exploration of very large interaction graphs, more complex architectures, or comprehensive sensitivity analyses.
- **External Validity:** Since experiments are conducted on specific datasets from particular platforms and industries, the findings may not generalise across all sectors, cultures, and regulatory environments. Results should be interpreted as most valid in contexts with comparable data, business structures, and customer engagement patterns.

# Chapter 2

## Literature Review

### 2.1 Preliminaries

This chapter introduces the key concepts, notation, and assumptions underlying the proposed Graph-Enhanced Causal Reinforcement Learning framework for prescriptive customer retention. It covers churn and retention economics, the transition from predictive to prescriptive analytics, causal uplift modelling and customer segmentation, offline reinforcement learning and batch-constrained contextual bandits, graph neural networks for representation and deconfounding, and off-policy evaluation with benchmark datasets. These preliminaries establish the formal building blocks for specifying the problem, deriving the learning objectives, and interpreting empirical results in subsequent chapters.

Customer churn denotes the termination of the relationship between a customer and a service provider within a defined period, while retention strategies aim to prevent this and sustain long-term relationships. Retention is strategically important because acquiring new customers is substantially more expensive than retaining existing ones, and small improvements in retention can yield significant financial gains in subscription-based settings. In such contexts, high churn erodes recurring revenue, and purely predictive churn models, though useful for ranking at-risk customers, offer limited guidance on which interventions will maximise incremental improvements in outcomes for particular customer profiles.

Traditional CRM systems treat churn management as a binary classification problem, training models such as logistic regression, random forests, or gradient-boosted trees to estimate the probability of churn. These models may achieve high accuracy but remain causally blind, capturing correlations without estimating how specific actions change customer behaviour. Consequently, a high churn score does not indicate whether an intervention will be effective, motivating methods that model causal effects and support prescriptive decision policies, such as uplift models and reinforcement learning-based CRM strategies.

Uplift modelling, viewed as Individual Treatment Effect (ITE) estimation, targets the causal impact of interventions by contrasting potential outcomes under treatment and control. Because only one outcome is observed for any realised treatment assignment, uplift models estimate the Conditional Average Treatment Effect (CATE), providing a group-conditional approximation of individual effects. Focusing on CATE enables policies that prioritise customers for whom treatment meaningfully changes outcomes, shifting the objective from predicting churn to maximising

causal uplift in retention or revenue.

CATE estimates enable segmentation into Persuadables, Sure Things, Lost Causes, and Sleeping Dogs, each reflecting distinct treatment responses. Persuadables benefit from treatment, Sure Things stay regardless, Lost Causes churn despite offers, and Sleeping Dogs exhibit negative uplift when contacted. Risk-based models that only estimate churn probability cannot separate these groups, leading to overspending on Sure Things and Lost Causes and unintentionally activating Sleeping Dogs, which reduces overall retention efficiency.

In high-stakes CRM environments, large-scale online exploration or naive A/B testing is costly and risky because actions directly affect customer relationships. Offline reinforcement learning instead learns decision policies from logged trajectories generated by a behaviour policy, without further environmental interaction. However, distributional shift can make value estimates unreliable for poorly supported actions, so offline RL algorithms introduce pessimism and constrain the learned policy to regions where the behaviour data are sufficiently informative.

Contextual bandits model decisions where an agent observes context, chooses an action, and receives a reward. In the batch-constrained setting, optimisation restricts the policy to actions with adequate empirical support in the logged data. Methods such as Batch-Constrained Q-Learning maximise the expected reward only over actions that are likely under the behaviour policy, often using a generative model to constrain the action space. Combined with uplift modelling, contextual bandits can focus on persuadable customers and optimise incremental causal effects rather than only immediate rewards.

Customer-item interactions are naturally represented as bipartite graphs with customers and items as nodes and observed interactions as edges. Graph Neural Networks learn node embeddings via message passing, exploiting collaborative structures and latent preferences not captured by tabular features. These graph-based state representations help mitigate sparsity and cold-start by sharing information across connected nodes and can partially deconfound the learning process, improving both uplift estimation and offline RL policy quality.

As counterfactual outcomes under unchosen actions are unobserved, off-policy evaluation (OPE) is required to estimate the performance of new policies using historical data from the behaviour policy. Techniques such as Inverse Propensity Scoring reweight outcomes by the ratio of target to behaviour policy probabilities, while Doubly Robust estimators combine outcome models with reweighting to reduce both bias and variance, providing a principled way to assess uplift-based and offline RL policies before deployment.

Realistic benchmark datasets are essential for evaluating prescriptive CRM models in a controlled yet representative manner. The Open Bandit Dataset provides large-scale logged interaction data with user context, actions, outcomes, and propensity scores from multiple behaviour policies, making it well suited for testing off-policy evaluation methods, uplift models, and contextual bandits. This dataset constitutes the primary empirical basis on which the proposed framework will be applied and evaluated.

## 2.2 Review of Existing Research

### 2.2.1 Customer Retention and Churn

Customer retention and churn prediction remain among the most impactful challenges in business analytics, directly affecting profitability and organisational sustainability. Classical churn models often prioritise classification accuracy while neglecting the economic heterogeneity of retention decisions. A profit-maximisation model that integrates Customer Lifetime Value estimation with retention campaign optimisation demonstrates that expected profit-lift targeting delivers superior returns compared with probability-based targeting, although it depends on accurate CLV estimates and the temporal stability of behavioural patterns [22]. Complementing this economic perspective, an ensemble reinforcement learning approach optimises intervention timing and content across the entire customer journey, showing how sequential optimisation over multiple touchpoints can significantly enhance conversion and retention, though large-scale experimental data from several journey stages are required [26].

A pivotal randomised field experiment reveals the phenomenon of retention futility, demonstrating that high-risk customers targeted with retention interventions often exhibit the lowest response rates [8]. This finding exposes a fundamental distinction between churn risk and intervention responsiveness, challenging the prevailing practice of prioritising high-risk customers and advocating instead for uplift-based strategies that focus on the anticipated causal effect of treatment. This paradigm shift directly motivates the causal inference component of the proposed framework. A complementary profit-driven churn management framework jointly optimises the target customer set and the retention offer, incorporating customer-specific profit calculations, variable intervention costs, and response probability estimation to achieve substantially higher profitability than traditional loss-function-based models [4].

In the telecom domain, a neural network-based predictive decision model simultaneously provides churn prediction and personalised retention recommendations based on domain-specific variables including usage patterns, billing records, and service contacts, achieving high predictive accuracy with practical relevance for managers, though its heavy dependence on domain-specific feature engineering limits cross-industry transferability [27]. A hybrid deep learning framework combining convolutional neural networks, long short-term memory networks, and fully connected layers captures both static customer features and dynamic behaviour patterns, achieving 95.4% accuracy on telecom churn data at the cost of model complexity, interpretability, and computational efficiency [33]. A comparative analysis of traditional machine learning and deep learning models for telecom churn prediction finds that gradient boosting algorithms achieve the best trade-off between accuracy and computational cost, and that effective feature engineering combined with class-imbalance handling through SMOTE-based oversampling is more critical for robust prediction than model complexity alone [34].

### 2.2.2 Causal Inference and Uplift Modeling

Personalised decision-making in marketing, healthcare, and policy relies on estimating heterogeneous treatment effects (HTE). Unlike average treatment effect estimation, which provides population-level information, HTE methods estimate



individual- or subgroup-level differences in response to treatment, enabling interventions targeted at those expected to benefit most. The meta-learner framework adapts standard machine learning algorithms to estimate conditional average treatment effects (CATE), comprising T-learner, S-learner, and X-learner variants; empirical analyses indicate that the X-learner performs best with imbalanced treatment groups, while the T-learner is preferred when sufficient data are available in both groups, though meta-learner selection substantially affects estimation accuracy and must be carefully validated [14].

DragonNet, a neural architecture that jointly estimates treatment effects and propensity scores through shared representations, uses the propensity head as a regulariser to reduce selection bias [15]. Generalized Random Forests (GRF) extend random forests with custom splitting criteria targeting treatment effect heterogeneity, providing theoretical guarantees for consistency and asymptotic normality along with valid confidence intervals through honest splitting [12]. Reformulating uplift modelling as a learning-to-rank task enables ordering individuals by expected treatment benefit to directly optimise ranking metrics such as the area under the uplift curve, thereby improving campaign efficiency for top-k targeting [29]. Counterfactual Regression (CFR) learns balanced representations by minimising the distribution distance between treatment and control groups while preserving predictive information, though overly aggressive balancing may discard outcome-relevant signals [5]. Extending these ideas to longitudinal data, causal contrastive learning for counterfactual regression over time uses contrastive learning to disentangle treatment-relevant and time-varying confounders for temporal treatment effect estimation [35].

Bayesian Personalized Ranking (BPR) is an optimisation criterion designed for implicit feedback scenarios that maximises the posterior probability that a user prefers observed items over unobserved ones using pairwise comparisons [2]. The BPR loss function is directly adopted as the training objective for LightGCN in the proposed framework, providing the theoretical foundation for learning user preference rankings from implicit interaction data.

### 2.2.3 Offline Reinforcement Learning and Bandits

Offline reinforcement learning (RL) learns optimal decision policies from pre-collected, static datasets without environmental interaction, which is particularly relevant in domains where online exploration is costly, risky, or impractical. Conservative Q-Learning (CQL) addresses extrapolation error through conservative regularisation that penalises overestimated Q-values for out-of-distribution actions, yielding substantially improved returns on suboptimal offline datasets at the cost of additional computation [21]. Implicit Q-Learning (IQL) uses expectile regression to estimate state values without explicit maximisation over actions, achieving strong performance with faster training but sensitivity to the expectile hyperparameter [30]. Batch-Constrained Q-Learning (BCQ) constrains the learned policy to actions well-supported by the behaviour data using a generative model for action selection and perturbation-based refinement, and serves as the core decision-making component in the proposed framework for its effective balance between policy conservatism and reward maximisation in batch-constrained settings [13].

The More Robust Doubly Robust (MRDR) estimator achieves minimum variance within the class of asymptotically optimal doubly robust methods by combining di-

rect predictions with importance sampling reweighting, and serves as the primary evaluation method in the proposed framework, enabling principled comparison of learned policies against baselines without live experimentation [9]. The Open Bandit Dataset and Pipeline provides a large-scale dataset of bandit feedback from the ZOZOTOWN e-commerce platform, containing logged data from random and Thompson Sampling behaviour policies with propensity scores; experiments demonstrate that the DRos estimator achieves 30–60% superior accuracy in OPE compared with conventional methods, and the dataset serves as the primary empirical basis for the proposed framework [28].

Causal reinforcement learning has been applied to subscription service churn prediction by integrating Granger causality for causal feature selection with RL-based policy optimisation, representing the closest existing approach to the proposed framework that combines causal reasoning with sequential decision-making for customer retention, though it operates without graph-based representations and does not address network confounding [36]. Formulating recommendation for long-term user engagement as an RL problem demonstrates that optimising for cumulative engagement rather than immediate click-through rates leads to substantially improved retention [19]. A comprehensive tutorial and survey of offline reinforcement learning synthesises fundamental challenges including distributional shift, value overestimation, and the conservatism–optimism trade-off, providing the theoretical grounding for the BCQ-based approach adopted in this research [23]. LinUCB, a contextual bandit algorithm for personalised recommendation, selects actions based on contextual features using linear reward models, establishing the foundational contextual bandit problem formulation that the proposed framework extends with neural Q-value estimation via BCQ [3].

#### 2.2.4 Core GNN Architectures

Graph neural networks (GNNs) have emerged as a powerful paradigm for learning representations on structured data, enabling applications in social networks, recommender systems, knowledge graphs, and molecular modelling. The Graph Convolutional Network (GCN) is the foundational spectral-based graph neural network that performs semi-supervised node classification using layer-wise propagation rules derived from a first-order approximation of spectral graph convolutions, establishing the message-passing paradigm upon which all subsequent GNN models build, including LightGCN, which is a direct simplification removing feature transformations and nonlinearities for recommendation tasks [7]. Graph Attention Networks (GAT) replace fixed neighbourhood aggregation with multi-head self-attention that assigns learnable attention weights to neighbouring nodes, achieving strong benchmark performance with interpretable attention weights at the cost of higher computation [10].

PinSage, one of the earliest industrial-scale GNNs for recommendation, combines random walk-based neighbourhood sampling with importance pooling to operate on billion-scale user–item graphs, demonstrating that GNN-based recommendation is feasible at web scale [11]. Neural Graph Collaborative Filtering (NGCF) learns high-order collaborative information through stacked graph convolutions with explicit feature transformation and interaction modelling at each propagation layer, serving as the direct predecessor to LightGCN [17]. LightGCN eliminates feature

transformation and nonlinear activation components, retaining only neighbourhood aggregation and layer-wise embedding combination; this radical simplification yields superior recommendation performance with fewer parameters and lower computational cost, making it the core GNN component in the proposed framework whose learned embeddings serve as dense, structurally informed state representations for the BCQ agent [20].

GraphSAGE, an inductive graph learning framework, generates embeddings for previously unseen nodes, directly supporting the cold-start argument for the proposed framework by enabling initial embedding generation for new customers with sparse interaction histories [6]. GNNExplainer, a model-agnostic framework for explaining GNN predictions, identifies influential subgraphs and feature subsets through mutual information maximisation, enhancing transparency in GNN-based decision-making, which is critical where retention decisions must be justifiable to business stakeholders [18].

### 2.2.5 GNN-Based Causal Inference

Causal inference in networked settings is complicated by interference, where one unit’s treatment affects others’ outcomes, violating the Stable Unit Treatment Value Assumption (SUTVA). Emerging methods combine graph neural networks with causal identification to address network confounding, peer effects, and treatment propagation. A network-embedding approach to causal inference demonstrates that nodes’ structural positions can serve as proxy variables for latent confounders that jointly affect treatment assignment and outcomes; by incorporating learned embeddings as covariates in causal models, bias from network-correlated unobservable confounders is reduced, providing the theoretical foundation for using LightGCN embeddings as deconfounded state representations in the proposed framework [16]. A formal GNN framework for causal inference under network confounding and interference, where outcomes depend on both a unit’s own treatment and the treatments and covariates of its neighbours, provides formal proof that GNN-learned representations improve causal effect estimation in networked settings, strengthening the theoretical justification for the proposed framework [40]. Synthesis of causal inference frameworks for network interference classifies interference patterns, reviews experimental designs such as cluster and graph-cluster randomisation, and demonstrates that network features such as hubs and community structure substantially shape causal conclusions [38]. An investigation of churn prediction under mutual customer interdependence demonstrates that features such as neighbours’ churn status, centrality, and community membership significantly enhance predictive accuracy on telecommunications data, supporting the view that churn exhibits socially contagious patterns; the proposed framework extends this work by integrating causal bandits for prescriptive intervention rather than purely predictive modelling [31].

### 2.2.6 Dynamic GNNs and Temporal Methods

Real-world networks evolve over time as nodes and edges appear, disappear, or change strength, reflecting shifting relationships and behaviours. Static graph models cannot adequately capture these non-stationary dynamics, motivating dynamic GNN approaches. TempODEGraphNet models churn in MMORPGs using time-

varying social graphs processed through GCN, LSTM, and Neural ODE components, achieving higher F1 scores than strong static baselines and demonstrating that dynamic representations capture evolving user behaviour more effectively [39]. Temporal Graph Networks (TGN), a continuous-time architecture that maintains node-level memory updated at each event and uses temporal attention over recent neighbours, achieves state-of-the-art performance on future edge prediction and dynamic node classification, though the framework remains purely predictive without causal or intervention analysis support [25].

EvolveGCN models temporal dynamics by evolving GCN weight matrices with recurrent units so that graph convolutions adapt to changing network structures without requiring persistent node identities, achieving competitive results on temporal link prediction tasks [24]. TSUNAMI, an explainable predictive process monitoring method for customer churn, combines temporal attention with multiple-instance learning to model variable-length customer histories, highlighting influential time periods and features while maintaining competitive predictive performance on retail datasets [37]. A comprehensive survey of dynamic graph neural networks distinguishes discrete-time snapshot-based models from continuous-time event-based models, noting that while DGNNs outperform static approaches, most research targets representation learning rather than causal inference or policy optimisation [32]. ConDyGNN, a context-aware dynamic GNN for developer churn prediction, combines gated temporal convolutions, diffusion convolutional layers, and multi-head attention over weekly collaboration graphs, achieving superior accuracy compared with ensemble and sequence-based baselines [41]. Although the proposed framework employs a static LightGCN for computational feasibility, these dynamic GNN approaches provide clear future extension paths for incorporating temporal evolution of customer interaction graphs.

## 2.3 Summary of Key Findings

The comprehensive review of related work reveals both substantial progress and critical gaps at the intersection of customer retention analytics, causal inference, offline reinforcement learning, and graph neural networks.

Traditional churn prediction models achieve increasingly high accuracy through deep learning architectures but remain fundamentally limited by their correlational nature. The retention futility phenomenon demonstrates that high churn risk does not imply high treatment responsiveness, invalidating the assumption underlying most CRM targeting systems. Uplift modelling provides the causal framework to identify Persuadable customers and avoid Sleeping Dog effects, but existing implementations operate without graph-based representations or sequential decision-making capabilities.

The offline RL literature demonstrates that optimal decision policies can be learned from logged data without costly online experimentation. Batch-Constrained Q-Learning effectively balances policy conservatism with reward maximisation in batch-constrained settings, while the Doubly Robust estimator and the Open Bandit Pipeline provide the evaluation infrastructure for reliable offline policy assessment. However, existing offline RL applications to customer retention do not incorporate graph-based state representations or explicit causal deconfounding.

GNN architectures from foundational graph convolutional networks through to the

simplified LightGCN demonstrate progressively effective methods for learning user-item interaction representations. LightGCN’s radical simplification yields superior recommendation performance with lower computational cost, making it the ideal GNN backbone for integration with offline RL. Inductive learning approaches provide the capability needed for cold-start scenarios, and GNN explainability frameworks enable the interpretability required for practical deployment.

The emerging literature on GNN-based causal inference establishes that graph embeddings can serve as proxy variables for unobserved confounders in networked settings. Research has demonstrated that network position encodes latent confounders and that GNN representations improve causal estimation under network confounding. These findings provide the theoretical justification for using graph embeddings as deconfounded state representations in a causal RL framework. Meanwhile, dynamic GNN approaches demonstrate that incorporating time-varying graph structure further improves churn prediction, offering clear future extension paths beyond the static graph modelling adopted in the initial framework design.

Despite significant advances across individual research streams, four critical gaps persist. First, customer retention, causal inference, offline RL, and GNN methods are developed independently with minimal cross-integration—no existing work combines all four components into a unified prescriptive retention framework. Second, new customers with sparse interaction histories remain difficult to represent effectively, and integrating cold-start GNN representations with causal bandit policies remains unexplored. Third, most offline RL and uplift modelling methods are validated on synthetic or curated benchmarks, and comprehensive evaluation on diverse industry datasets is lacking. Fourth, the Sleeping Dog phenomenon—customers who churn because of intervention—is recognised conceptually but lacks explicit algorithmic safeguards in existing RL or bandit frameworks. These gaps collectively motivate the proposed Graph-Enhanced Causal Reinforcement Learning framework, which integrates graph embeddings as deconfounded state representations, causal uplift estimation for treatment effect targeting, Batch-Constrained Q-Learning for safe offline policy learning, and Doubly Robust Off-Policy Evaluation for principled validation.

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